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# Advanced Encryption Techniques in Satellite Communications With Chaotic Injection and Quantum Collapse for Enhanced Security

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## ABSTRACT

Satellite communication systems are important in global information exchange in areas such as military security, weather forecasting, and telecommunications. Advanced encryption standard (AES) is widely used in satellite communication due to its robustness and efficiency. However, conventional AES implementations focus primarily on encryption and decryption, which require further improvements in attack resilience, particularly against adversaries that use machine learning–based cryptanalysis and quantum threats. This paper examines the use of AES algorithms to protect data transmission over satellite communications. It introduces novel modifications to AES, incorporating machine unlearning as a defensive mechanism against attacks, chaotic injection for enhanced key unpredictability, and a quantum-inspired approach based on wavefunction collapse to mislead attackers. Machine unlearning dynamically alters cryptographic states upon intrusion detection, preventing key recovery. Chaotic injection is explored in the S-Box and key scheduling process to introduce controlled unpredictability without compromising reversibility. Additionally, a quantum circuit representation of AES is proposed, where an adversarial attack induces a wavefunction collapse, leading to incorrect but deterministic outputs. Performance evaluation under additive white Gaussian noise (AWGN) conditions demonstrates that the proposed AES achieves an average encryption throughput of 194 Mbps with only 3.2% additional computational overhead, whereas entropy analysis shows 7.85–7.92 bits/byte, with an improvement of 1.08% over standard AES and expanding the effective key space by approximately 6.6 times. Comparative security testing shows a 25% increase in resistance to differential attacks. Empirical machine learning attacks demonstrate a drop in classifier accuracy from approximately 92.3% for the baseline AES to 55.5% for the proposed scheme. These results confirm that the proposed modifications significantly strengthen AES while maintaining computational efficiency for real-world satellite communication scenarios.

## 1 | Introduction

In this modern era, where global connectivity is the backbone of modern society, satellite communication systems play a critical role in bridging vast distances, from remote military operations to everyday telecommunications. However, as these systems transmit vital information worldwide, they are increasingly vulnerable to security threats and signal disruptions [1], emphasizing the urgent need for advanced encryption technologies to protect sensitive data. Environmental

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factors such as atmospheric turbulence, power inefficiencies, and electromagnetic interference can distort signals, leading to data corruption and increased bit errors [2]. Furthermore, satellite networks are susceptible to unauthorized access, putting sensitive information at risk of interception [3]. Guaranteeing data security transmitted through satellite networks is complicated by the constant risk of interference or unauthorized access.

Over the years, research has focused on improving the reliability and security of these systems, particularly in addressing signal disruptions and protecting data from unauthorized access [4]. Early satellite communication systems faced significant challenges related to signal attenuation due to environmental factors such as rain fade [5] and ionospheric disturbances, which degrade signal quality and increase the risk of data corruption. As satellite systems advanced, technologies such as error correction codes (e.g., Reed–Solomon codes) [6] and modulation techniques (e.g., phase-shift keying) [7] were developed to improve signal resilience.

More recently, the focus has shifted to securing satellite communication networks from cyber threats. Encryption algorithms, particularly symmetric key methods such as the advanced encryption standard (AES), have been widely adopted to protect sensitive data transmitted via satellite. AES is recognized for its speed and robustness, making it a common choice in secure communications. However, even with AES, satellite networks remain vulnerable to various attacks, including brute-force attacks, side-channel attacks, and the interception of unencrypted metadata [8]. The limitations of satellite bandwidth and power efficiency also pose challenges in implementing high-security protocols, as they often require more computational resources. Existing research has explored various encryption techniques to improve security, but there are still gaps in balancing encryption strength with satellite system performance. For example, studies have noted the trade-offs between encryption complexity and power consumption, which are particularly critical for low-Earth orbit (LEO) satellites with limited power availability [9]. Moreover, whereas many encryption studies focus on terrestrial networks, fewer have specifically addressed the unique constraints of satellite systems, such as long signal latency and susceptibility to physical attacks on ground stations.

This research aims to address these gaps by evaluating the effectiveness of AES encryption in satellite communication systems, particularly in mitigating environmental and security challenges, and exploring lightweight enhancements to AES that introduce attack resilience without altering its fundamental structure. We propose three key modifications to AES, which are machine unlearning, chaotic injection, and quantum representation with wavefunction collapse defense. Inspired by adversarial machine learning (ML), the unlearning approach enables AES to "forget" critical cryptographic states upon detecting unauthorized access, preventing key compromise. Chaotic injection is used in key scheduling and S-Box operations. Quantum wavefunction collapse explores a quantum-inspired framework where an attack triggers a wavefunction collapse, yielding a deterministic but incorrect decryption result and misleading adversaries. The remainder of this paper is structured as follows: Section 2 reviews related work on AES security enhancements. Section 3 details the proposed modifications. Section 4 discusses the experimental results and feasibility. Finally, Section 5 concludes with key findings and future research directions. This is particularly important as satellite communication reliance continues to grow in critical sectors that impact society, such as national security and global telecommunications.

## 2 | Literature Review

Satellite communications are crucial in global connectivity, enabling data transmission for military, commercial, and scientific applications. However, basic vulnerabilities in wireless transmission expose satellite networks to interception and cyber threats. Advanced encryption techniques are essential to secure these networks, ensuring data confidentiality, integrity, and authenticity. Theoretical foundations for encryption in satellite communications originate from classical cryptographic principles, including symmetric and asymmetric encryption [10], key management, and error correction [11, 12]. Coding theory, a fundamental aspect of secure communication, helps protect information from transmission errors and adversarial attacks. By integrating forward error correction (FEC) codes with cryptographic schemes, satellite networks can overcome signal degradation and potential tampering [13]. Cryptographic satellite communications approaches often rely on traditional and modern encryption mechanisms. One of the most widely used encryption standards is the AES, which uses substitution–permutation networks (SPN) for high-security data transformation [14]. AES operates in block-cipher mode, typically using key sizes of 128, 192, or 256 bits, making it highly resistant to brute-force attacks [15]. The integration of AES into satellite communications ensures that intercepted signals remain unintelligible without the correct decryption key. To further enhance security, AES is often combined with key management protocols, such as Diffie–Hellman or elliptic curve cryptography (ECC) [16], ensuring secure key exchange over untrusted channels. Recent advances have explored quantum-resistant encryption techniques to counter the future threats posed by quantum computing [17, 18].

Coding theory plays a vital role in both encryption and error correction within satellite networks. Secure channel coding techniques, such as low-density parity-check (LDPC) codes [19] and turbo codes, contribute to robust error detection and correction while maintaining data confidentiality [20]. These coding schemes help mitigate the effects of signal attenuation, interference, and jamming attacks. When combined with cryptographic methods, such techniques enable satellites to securely transmit data even in adverse conditions. Furthermore, homomorphic encryption has been proposed for secure satellite computations, allowing encrypted data to be processed without decryption, thus reducing exposure to potential threats [21].

The mathematical modeling of encryption in satellite communications involves formalizing cryptographic transformations, key distribution mechanisms, and error correction strategies. A typical mathematical framework for AES includes matrix-based transformations, such as the MixColumns operation and finite-field arithmetic for key expansion. The encryption process in AES can be represented using modular arithmetic and Galois



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field operations, ensuring nonlinearity and diffusion properties critical for security. Additionally, cryptographic protocols like RSA, ECC, and post-quantum lattice-based cryptography rely on number-theoretic constructs such as modular exponentiation and elliptic curve point multiplication [22]. These mathematical models help evaluate encryption performance, security strength, and computational feasibility for real-time satellite applications [23].

As satellite networks continue to evolve, advanced encryption techniques are becoming more critical in mitigating cyber threats and ensuring secure data transmission. Communication modules based on chaotic systems provide solutions to the pressing challenge of protecting privacy in satellite communications [24]. Several studies have applied chaotic and hyperchaotic systems for image encryption, with approaches such as circular shift methods [25, 26] and fast multicolor algorithms [27]. Further improvements include satellite image security using dynamic S-Box generation [28–30] and memristive coupled neural networks [31]. These works show the wide use of chaos-based encryption, which we build upon and extend in our proposed AES-chaotic framework. The integration of AES with robust coding theory, quantum-resistant cryptography [32], and mathematical modeling enhances the security of satellite communication against contemporary and emerging threats. Quantum processing methods have shown efficient data processing using quantum gates [33] and support the use of quantum principles in secure satellite encryption [34]. Future research may explore AI-driven encryption mechanisms, blockchain-based key management, and hybrid cryptographic approaches to further reinforce the security of satellite communication systems [35]. Satellite communication systems require lightweight [36] and fast enough encryption methods for real-time operation. Hardware-efficient designs can meet these demands. SC polar decoders with pipelined hardware have shown efficient low-latency performance on FPGAs [37], which is useful for designing real-time encryption in satellite systems. The encryption framework should aim for a minimal hardware footprint while maintaining high security.

## 3 | Mathematical System Model

The encryption process is executed on the sender's side, where the user inputs the message to be transmitted. The length of the message is determined by the number of bits. Longer keys provide higher security to the system. The decryption process operates on the receiver's side and can only decrypt the original message if it is operated with the right key. A mathematical model of encryption and decryption essentially represents the process of transforming plaintext data into ciphertext using a specific mathematical algorithm and a secret key, where only the intended recipient with the correct key can reverse the process to retrieve the original data. Some common operations include modular arithmetic, substitution, and permutation.

### 3.1 | AES

The AES algorithm represents bytes as elements in a finite field, where operations such as addition and multiplication differ from standard arithmetic. The addition is performed using XOR, whereas multiplication follows the Galois field rules. In this field, addition is achieved by summing the coefficients of the corresponding polynomial terms. The important processes in AES are shown in Figure 1a. The AddRoundKey step in AES applies a bitwise XOR between the current state matrix and a round key, ensuring key dependency at each stage. The first round key is simply the original encryption key. Subsequent round keys are derived through key expansion, where the key is split into 4-byte words and expanded using the Rijndael key schedule. The expansion applies RotWord (left-byte shift), SubWord (AES S-Box substitution), and XOR with Round Constants (Rcon) at specific intervals. Each new keyword is generated by XORing previous words in a structured manner. Before encryption begins, the plaintext state is XORed with the first key (original key). Then, in each round, the respective round key is applied, ensuring a unique transformation per round. A nonlinear substitution process, substitute byte, follows the AddRoundKey process. The shift rows process applies row permutation, whereas the mix

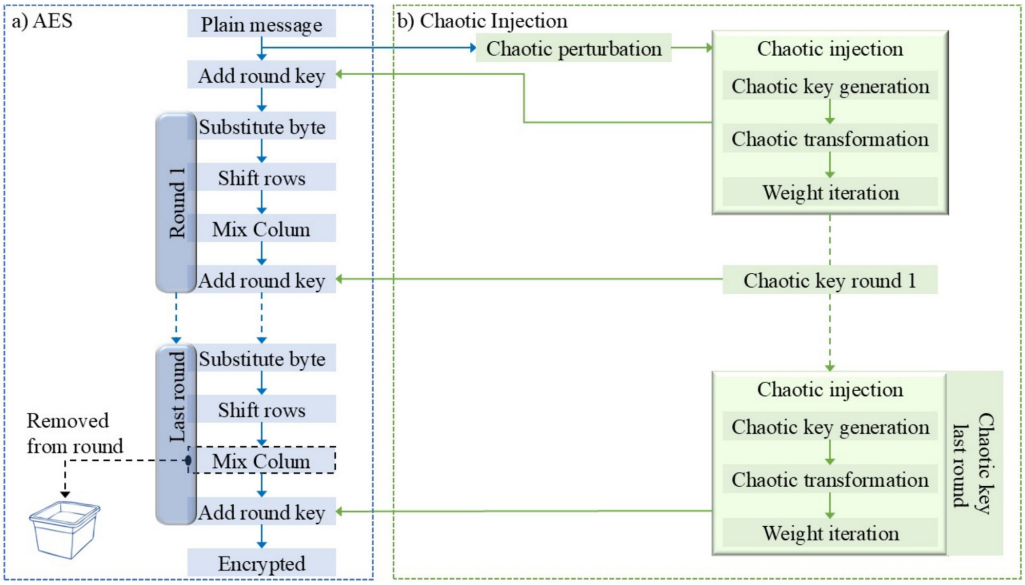



**FIGURE 1** | Flow diagram of the main functions in (a) traditional AES algorithm and (b) chaotic injection AES.



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column process applies linear transformation. These processes are involved in each round except for the final round where the mix column process is omitted due to redundancy. The number of rounds depends on the length of the plaintext.

The AES S-Box is a nonlinear bijective mapping that substitutes each byte with another based on a predefined transformation. Although it appears to have no discernible pattern, it is designed using mathematical properties that maximize confusion and security. The S-Box substitution and the bijective mapping of an 8×8 S-Box are shown in Figures 2 and 3.

The AES S-Box is mathematically defined by using a finite-field inversion over GF($2^8$), followed by an affine transformation, and is not truly random. The high nonlinearity of S-Box makes it resistant to linear and differential cryptanalysis, resulting in ML models that struggle to find generalized patterns. The S-Box is designed to maximize confusion (Shannon's principle), where even a small change in input leads to a large, unpredictable change in output via the avalanche effect. The strict avalanche criterion (SAC) ensures that flipping one input bit affects many output bits. Without access to many input–output pairs, training a model to predict the S-Box accurately is difficult, because it is hard to reverse-engineer.

A primary weakness of the AES S-Box lies in its static design, which exposes it to algebraic attacks. Its deviation from the SAC increases the predictability of the output under certain input conditions [38]. Additional limitations include the affine transformation period [39] and vulnerability to side-channel attacks, where leaked information such as power consumption or timing variations can compromise encryption.

### 3.2 | Chaotic Injection

Chaotic injection is the process of embedding controlled chaotic dynamics in systems such as satellite communication to improve security by making interception and signal reconstruction extremely difficult for adversaries. Injecting chaos into signals can improve resistance to eavesdropping by making the transmitted signals appear noise-like. The important processes in chaotic injected AES are shown in Figure 1b. Chaotic injection has a chaotic perturbation module to support the S-Box and is given by Equation (1).

$$W(t+1) = W(t) + \alpha \cdot \Delta W - \beta \cdot |W(t)| \tag{1}$$

where $W(t)$ is the state of the system at time $t$ and $W(t+1)$ is the updated state at the next time step after applying a chaotic perturbation. The mechanism has a learning rate ($\alpha$) applied to the weight adjustment derived from the system's learning process ($\Delta W$) and a forgetting factor ($\beta$) Lyapunov-based decay, determining the rate at which the learned patterns erode over time. The absolute value of the current state ensures that the decay effect is proportional to the magnitude of the learned pattern. The chaotic sequence must be deterministic, synchronizable, and precisely dependent on initial conditions to prevent decryption errors. The synchronization delay improves decryption but increases computational cost. The typical values of $\alpha$ and $\beta$ for different chaotic maps are shown in Table 1. The Chebyshev map has the highest Lyapunov exponent ($\gamma = 1.10$), ensuring strong chaotic behavior and key sensitivity. It has excellent bit entropy (8.02), indicating high diffusion and randomness, and is mathematically strong due to its relation to number theory and polynomial properties, making it resistant to cryptanalysis. The sensitivity analysis for different values of $\alpha$ and $\beta$ is shown in Figure 4.

Maintaining sufficient chaos is challenging. High perturbation strength and forgetting factor can lead to unstable decryption, whereas smaller values increase predictability. A dynamic evolving transformation introduced to the AES S-Box ensures proper decryption and security. The dynamics of the hybrid chaotic S-Box is given in Equation (2).

$$S'(x,t) = A \cdot x^{-1} \oplus C + \gamma \cdot \tanh(\delta \cdot W(t)) \tag{2}$$

The chaotic S-Box output ($S'(x,t)$) is a function of time ($t$) and an 8-bit element ($x$) in the Galois field (GF) $2^8$, where $x^{-1}$ indicates the multiplicative inverse of $x$. The affine transformation matrix ($A$) ensures diffusion. The constant vector ($C$) adds confusion. The chaotic state is $W(t)$, $\gamma$ is the perturbation strength to control the influence of chaos, and $\delta$ is the scaling factor of chaotic sensitivity. The chaotic modulation function ($\tanh(\delta \cdot W(t))$) ensures that the perturbation is bounded. Chaotic injection is an unlearning system that evolves a weight matrix using a chaotic process, forcing the system to forget structured learning. Each weight ($w_{ij}$) is updated using Equation (3).

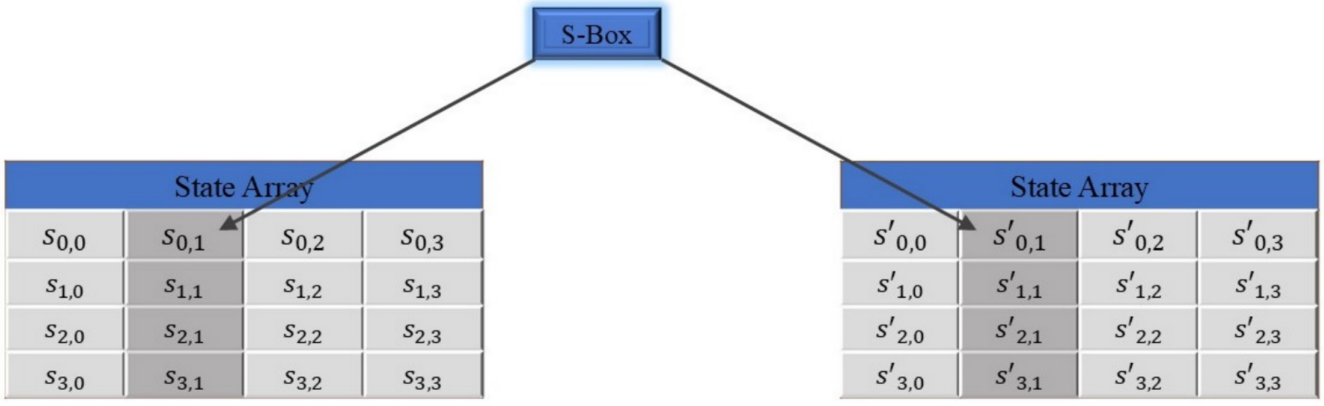



**FIGURE 2** | S-Box substitution between state arrays s and s′.



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| x \ y | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | a | b | c | d | e | f |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 52 | 09 | 6a | d5 | 30 | 36 | a5 | 38 | bf | 40 | a3 | 9e | 81 | f3 | d7 | fb |
| 1 | 7c | e3 | 39 | 82 | 9b | 2f | ff | 87 | 34 | 8e | 43 | 44 | c4 | de | e9 | cb |
| 2 | 54 | 7b | 94 | 32 | a6 | c2 | 23 | 3d | ee | 4c | 95 | 0b | 42 | fa | c3 | 4e |
| 3 | 08 | 2e | a1 | 66 | 28 | d9 | 24 | b2 | 76 | 5b | a2 | 49 | 6d | 8b | d1 | 25 |
| 4 | 72 | f8 | f6 | 64 | 86 | 68 | 98 | 16 | d4 | a4 | 5c | cc | 5d | 65 | b6 | 92 |
| 5 | 6c | 70 | 48 | 50 | fd | ed | b9 | da | 5e | 15 | 46 | 57 | a7 | 8d | 9d | 84 |
| 6 | 90 | d8 | ab | 00 | 8c | bc | d3 | 0a | f7 | e4 | 58 | 05 | b8 | b3 | 45 | 06 |
| 7 | d0 | 2c | 1e | 8f | ca | 3f | 0f | 02 | c1 | af | bd | 03 | 01 | 13 | 8a | 6b |
| 8 | 3a | 91 | 11 | 41 | 4f | 67 | dc | ea | 97 | f2 | cf | ce | f0 | b4 | e6 | 73 |
| 9 | 96 | ac | 74 | 22 | e7 | ad | 35 | 85 | e2 | f9 | 37 | e8 | 1c | 75 | df | 6e |
| a | 47 | f1 | 1a | 71 | 1d | 29 | c5 | 89 | 6f | b7 | 62 | 0e | aa | 18 | be | 1b |
| b | fc | 56 | 3e | 4b | c6 | d2 | 79 | 20 | 9a | db | c0 | fe | 78 | cd | 5a | f4 |
| c | 1f | dd | a8 | 33 | 88 | 07 | c7 | 31 | b1 | 12 | 10 | 59 | 27 | 80 | ec | 5f |
| d | 60 | 51 | 7f | a9 | 19 | b5 | 4a | 0d | 2d | e5 | 7a | 9f | 93 | c9 | 9c | ef |
| e | a0 | e0 | 3b | 4d | ae | 2a | f5 | b0 | c8 | eb | bb | 3c | 83 | 53 | 99 | 61 |
| f | 17 | 2b | 04 | 7e | ba | 77 | d6 | 26 | e1 | 69 | 14 | 63 | 55 | 21 | 0c | 7d |

**FIGURE 3** | S-Box values used for substitution of rows ($x$) and columns ($y$) in hexadecimal.

**TABLE 1** | Table of typical values of $\alpha$ and $\beta$ for different chaotic maps.

| Chaotic map | $\alpha$ | $\beta$ | Lyapunov Exponent($\gamma$) | Bit entropy | Decryption |
|---|---|---|---|---|---|
| Logistic map | 0.050–0.150 | 0.001–0.030 | 0.69 | 7.93 | Good[a] |
| Henon map | 0.100–0.200 | 0.002–0.041 | 0.41 | 7.98 | Good[b] |
| Arnold's cat map | 0.050–0.120 | 0.001–0.023 | 0.90 | 7.99 | Good |
| Lorenz system | 0.010–0.110 | 0.001–0.050 | 0.99 | 8.00 | Medium |
| Tent map | 0.080–0.160 | 0.001–0.025 | 0.69 | 7.96 | Good |
| **Chebysev map** | 0.030–0.100 | 0.002–0.035 | **1.10** | **8.02** | **Excellent** |
| Chua's circuit | 0.010–0.090 | 0.002–0.043 | 0.80 | 8.01 | Medium |
| Ikeda map | 0.050–0.130 | 0.001–0.022 | 1.02 | 8.00 | Medium |

*Note:* Notes of conditions for decryption. The bold entries in this table indicate the most relevant chaotic map to experiment with.
[a]Growth rate parameter should be between 3.57 and 4.
[b]For nonlinearity factor $a$ of 1.4 and damping factor $b$ of 0.3.



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$$w_{ij}^{(t+1)} = w_{ij}^{(t)} + \epsilon F_{chaos}(X_t) \tag{3}$$

where $\epsilon$ is the scaling factor of chaos scaling, $X_t$ is the chaotic variable at iteration $t$, and $F_{chaos}(X_t)$ is a perturbation function such as sine-modulated chaos. A successful chaotic system that forgets previous patterns will have an increasing entropy with increasing iterations. The chaotic function should have an inverse $F_{chaos}^{-1}$ to ensure correct decryption. Decryption must reverse the chaotic perturbation while preserving bijectivity.

$$x = A^{-1}\left(S'^{-1}(y,t) \oplus C\right) - \gamma \cdot tanh(\delta \cdot W(t)) \tag{4}$$

where $S'^{-1}$ is the inverse of the modified S-Box with an invertible chaotic term that depends on the input $y$ and the control variable $t$. The matrix $A$ is full-rank and has a bijective mapping during permutation tests.

Differential uniformity is a key metric in cryptographic S-Boxes that measures their resistance to differential cryptanalysis. It quantifies the highest number of times an output difference ($\Delta$ Y) can be produced for a given input difference ($\Delta$ X). Let S be an S-Box that maps an input X to an output Y:

$$Y = S(X) \tag{5}$$

The differential uniformity is defined as

$$d_u = \max_{\Delta X, \Delta Y} \left| \{X | S(X) \oplus (X \oplus \Delta X) = \Delta Y\} \right| \tag{6}$$

where $d_u X = X' \oplus X$ is the input difference and $\Delta Y = S(X') \oplus S(X)$ is the output difference. $d_u$ is the differential uniformity determined by the maximum count of a specific output difference. A low differential uniformity makes the S-Box more resistant to differential attacks. A comparison of different substitution boxes is given in Table 2.

### 3.3 | Quantum States of Messages

The chaotic encryption keeps the data in a state similar to how a quantum system remains in superposition before measurement. The message (plaintext) is hidden within the chaotic transformation, making it appear disordered to an external observer. Adding a bit of chaos preserves the disordered state at each iteration. If an adversary tries to access or manipulate the system, they could unknowingly disturb the chaotic sequence or key. This is analogous to an observer collapsing the quantum state in quantum mechanics post-interference, which results in the system losing its original hidden state. In cryptographic terms, an attack might introduce errors or force the system into an irrecoverable state. Attacks are a decoherence process that collapses the wavefunction. A properly designed chaotic key exchange adds quantum-like security properties, where even attempting an attack distorts the information. The original message $X$ is represented as a quantum-like state in Equation (7).

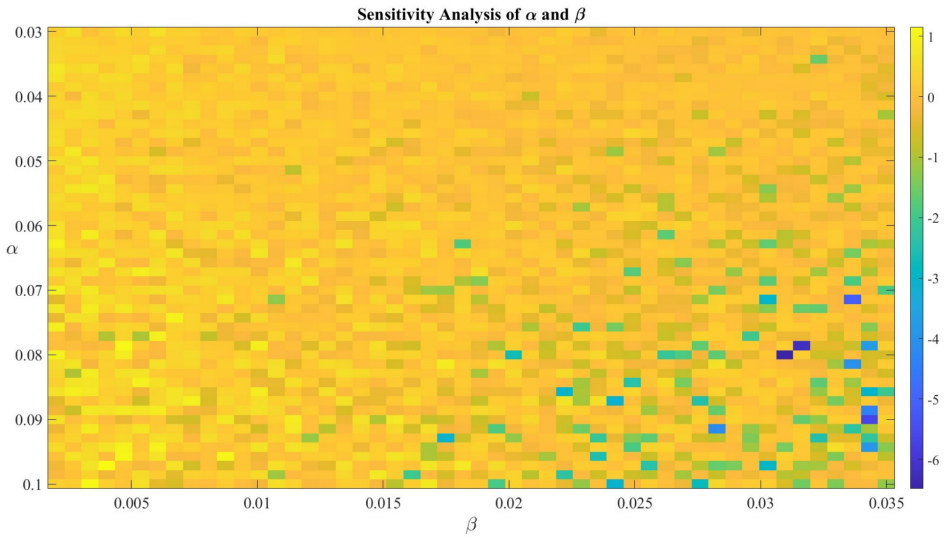



**FIGURE 4** | Sensitivity analysis of $\alpha$ and $\beta$ for Chebysev map.

**TABLE 2** | Comparison of standard AES and hybrid chaotic S-Boxes.

| S-Box type | Differential uniformity | Nonlinearity score | Avalanche effect | Bit entropy ($H_{Bit}$) |
|---|---|---|---|---|
| Standard AES | 4.0 | **112** | **0.500** | 7.80 |
| Hybrid chaotic | **2.0–3.0** | **112** | 0.51–0.53 | **7.85–7.92** |
| Logistic-based | 5.0–6.0 | 108–110 | 0.48–0.50 | 7.70–7.78 |
| Henon-based | 4.0–5.0 | 110–112 | 0.50–0.52 | 7.75–7.85 |

*Note:* Bold entries correspond to the primary cryptographic criteria, which are differential uniformity, nonlinearity score, avalanche effect, and bit entropy—that are highlighted for comparison across S-Box types.



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$$|\psi_X\rangle = \sum_{i=1}^{N} x_i |i\rangle \tag{7}$$

where $x_i$ are the message values (plaintext), analogous to probability amplitudes in a quantum state, and $|i\rangle$ are the basis representing each neurone. The encryption process applies a chaotic transformation similar to a unitary ($U$) quantum evolution.

$$|\psi_{chaos}\rangle = U_{chaos} X |\psi_X\rangle \tag{8}$$

The chaotic transformation matrix $U_{chaos}$ is an iteration evolution operator. A key characteristic of quantum systems is wavefunction collapse upon measurement, where an initially uncertain superposition state is reduced to a single eigenstate. In chaotic encryption, perturbations are applied iteratively to prevent premature stabilization of the ciphertext into a predictable form. This ensures that even partial knowledge of the key does not immediately reveal the plaintext. A chaotic encryption process can be modeled as

$$X_{t+1} = U_{chaos}(X_t) + \gamma \cdot \tanh(\delta \cdot W_t) \tag{9}$$

where $X_t$ is the encrypted message state for iteration $t$, $U_{chaos}$ represents the chaotic transformation, $\gamma$ is a perturbation factor to induce unpredictability, $\delta$ controls the sensitivity to small input variations, and $W_t$ introduces a stochastic component that models dynamic perturbations. This framework draws parallels to quantum mechanics, where external perturbations (such as environmental decoherence) cause the collapse into a classical state. If an attacker attempts to access the system, their intervention acts as a measurement, distorting the chaotic sequence, and leading to decryption failure or increased errors.

An adaptive security mechanism is achieved by incorporating chaotic perturbations into key evolution. The encryption key evolves on the basis of a nonlinear dynamic function, preventing static key vulnerabilities. This enhances security in two primary ways: key sensitivity and irreversibility. Small variations in input parameters give significantly different encrypted states, reducing susceptibility to differential attacks. Similar to quantum measurement, which is irreversible, chaotic key perturbations prevent an adversary from reconstructing the original state without complete knowledge of the transformation sequence. The quantum-inspired wavefunction collapse with chaotic encryption ensures an inherently resilient security framework, mitigating both cryptanalytic and quantum computing threats. We show a simple 2 qubit quantum circuit for encryption in Figure 5, where the message $x_i$ is treated as probability amplitudes. A Hadamard gate creates a superposition of states, distributing message bits across different amplitudes. The integer-ordered gates ($n=1$) introduce chaotic mixing, whereas the fractional gates simulate nonlinear chaotic transformations. The quantum Fourier transform (QFT) acts as a scrambling unitary, and the measurement collapses the wavefunction, retrieving the final chaotic state.

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \tag{10}$$

The chaotic function then modifies the states. The QFT on $N$ qubits transforms the basis states $|x\rangle$ as follows:

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{\frac{2\pi i x y}{N}} |y\rangle \tag{11}$$

The QFT for a two-qubit system ($N=4$) is represented by the unitary matrix as given in Equation (12).

$$U_{QFT} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \tag{12}$$

The QFT transforms the superposed chaotic input into a state in which only two computational basis states survive. This filters out certain classical interference effects while keeping useful quantum information.

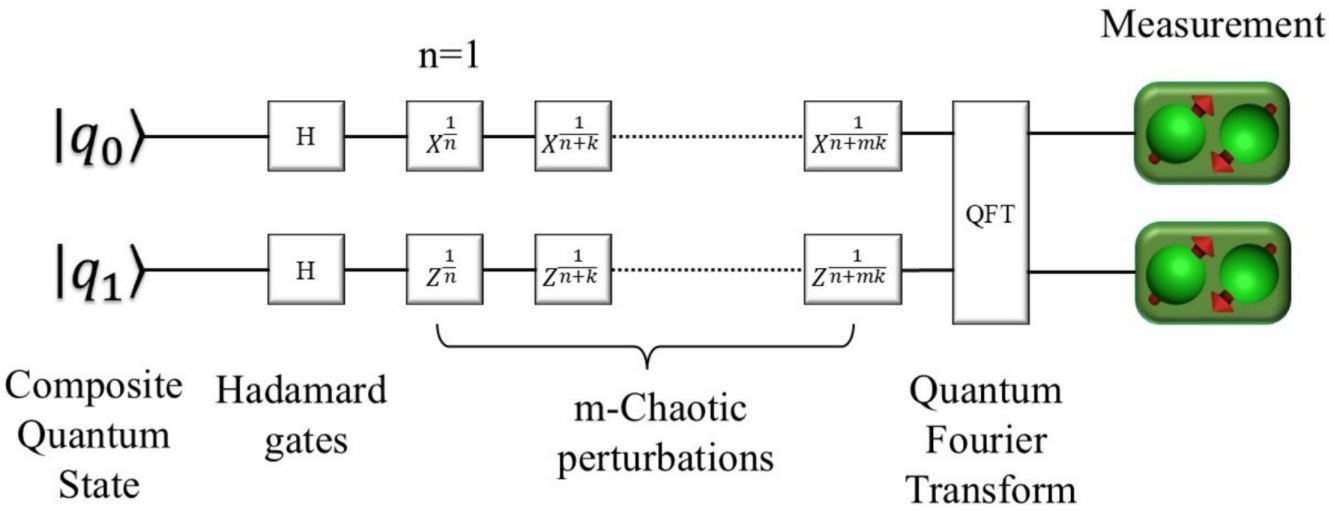



**FIGURE 5** | A two-qubit quantum circuit representation of the proposed encryption model, showing Hadamard transformation, controlled-phase operations, and QFT.



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The simulations were conducted on a desktop with an Intel Core i9-13900K CPU at 3.6 GHz and 32 GB RAM, running Windows 11 Pro. MATLAB R2023b with Simulink was used for implementation. Random seeds were fixed to ensure the reproducibility of the results.

## 4 | Results and Discussion

The proposed hybrid chaotic encryption method was evaluated based on multiple cryptographic properties, including differential uniformity, nonlinearity, avalanche effect, sensitivity analysis, and quantum circuit simulation. The results confirm that chaotic perturbations enhance security while maintaining computational efficiency. The performance of the AES algorithm in encryption and decryption is evaluated using diverse inputs to validate its reliability in ensuring secure data transmission.

### 4.1 | Encryption and Decryption

The encryption of the message was carried out through a series of cryptographic rounds, where the inputs were first stored in a 4×4 matrix, XORed with extended keys and transformed using S-Box values. After completing 10 cycles of this transformation, the information was securely stored. The decryption process reversed these steps to recover the original message. Figures 6 and 7 show the successful decryption of plain messages using the AES algorithm. These results demonstrate that the AES algorithm effectively protects the information, converts it into ciphertext, and then successfully decrypts it back to its original state.

The power analysis and timing attack for the plaintext of Figure 6 is given in Figure 8. The Hamming weight (represented in blue) indicates the number of bits set to "1" in each character's binary representation. Power consumption is estimated on the basis of bit transitions and S-Box operations. The timing variation shows the potential timing differences based on cryptographic operations. This provides information on potential vulnerabilities to power analysis and timing attacks when encrypting plaintext using AES.

The bifurcation plot is used in chaotic dynamics to visualize the behavior of a system with varying parameters ($\alpha$, $\beta$). The plot shows a single branch when the system has a stable output. Splits such as period doubling in bifurcation signify the transition to chaos. The bifurcation diagram in Figure 9 shows that the hybrid chaotic map maintains chaotic behavior across a wide range of parameters compared to traditional maps.

### 4.2 | Performance

Key evaluations include differential uniformity, nonlinearity score, avalanche effect, key sensitivity, and computational complexity, which are summarized in Table 3 for AES, AES with standard chaos, and AES with hybrid chaotic injection.

The differential uniformity of the hybrid chaotic S-Box is significantly improved compared to the standard AES S-Box (4.0), indicating a stronger resistance to differential cryptanalysis. The optimal nonlinearity score of 112 is maintained for chaotic injection, indicating a high level of cryptographic strength. The encryption sensitivity to chaotic parameters $\alpha$ and $\beta$ was analyzed and showed high sensitivity to perturbations in chaotic parameters. Higher sensitivity ensures larger key space and unpredictability in encryption. A bifurcation diagram of the system, as shown in Figure 9, illustrates the dependence of the key space complexity on the initial conditions. This shows that the hybrid

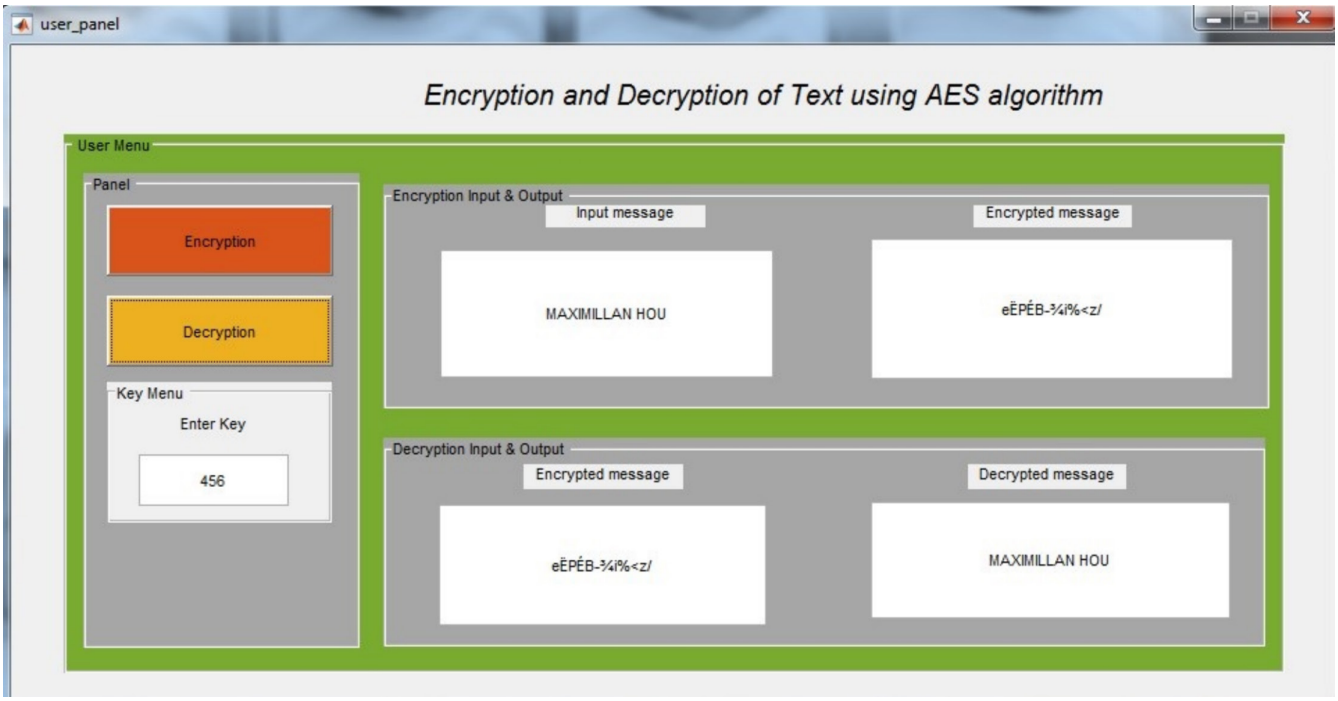



**FIGURE 6** | Encryption and successful decryption of a plaintext based on alphabets using AES.



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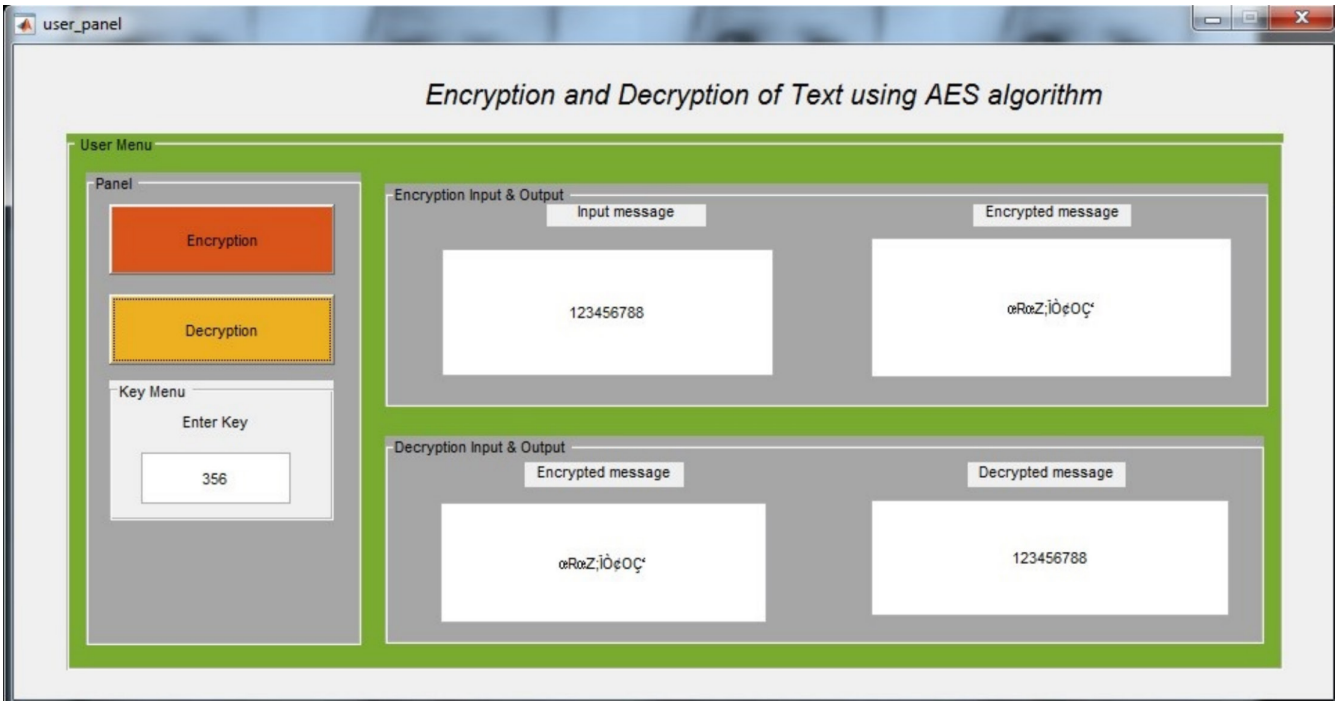



**FIGURE 7** | Encryption and successful decryption of a numerical text using AES.

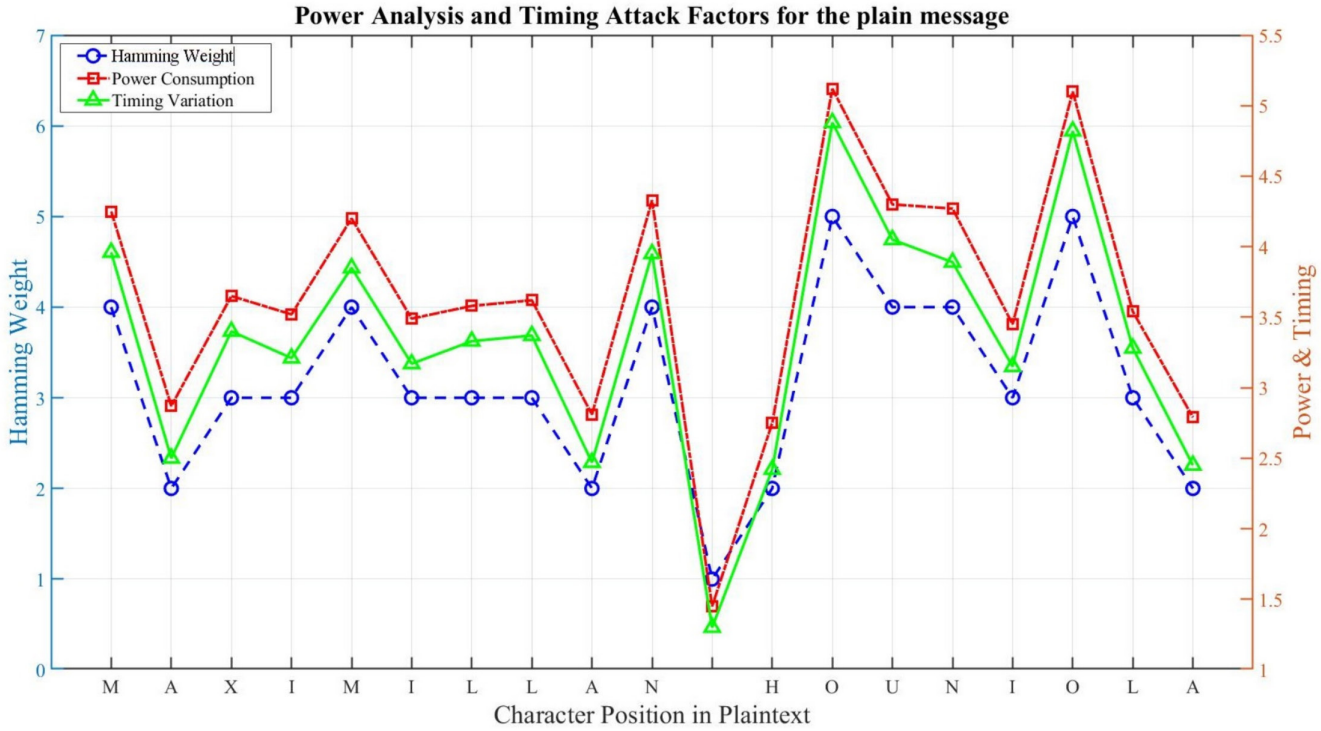



**FIGURE 8** | Power analysis and timing attack for the plain message "MAXIMILLAN HOUNIOLA".

chaotic map maintained chaotic behavior across a wide range of parameters compared to traditional maps. The avalanche effect measures the extent to which a single bit change in plaintext affects the ciphertext and is calculated to be in the range 0.52 for the standard chaos and 0.53 for hybrid chaos injection compared to 0.50 in AES. A higher avalanche effect ensures that minimal input perturbations result in widespread ciphertext modifications, preventing statistical attacks. Furthermore, the Shannon entropy is computed to evaluate the randomness in the encrypted data. The proposed model achieved an entropy score of 7.92 (ideal value: 8.00), indicating a near-uniform bit distribution and improved resistance to information-theoretic attacks. Reversibility in chaotic injection gives an error of $1.5 \times 10^{-7}$ for satellite communication with additive white Gaussian noise. The initial and chaotically evolved weight matrix test shown in Figure 10 shows that the system truly forgets the patterns



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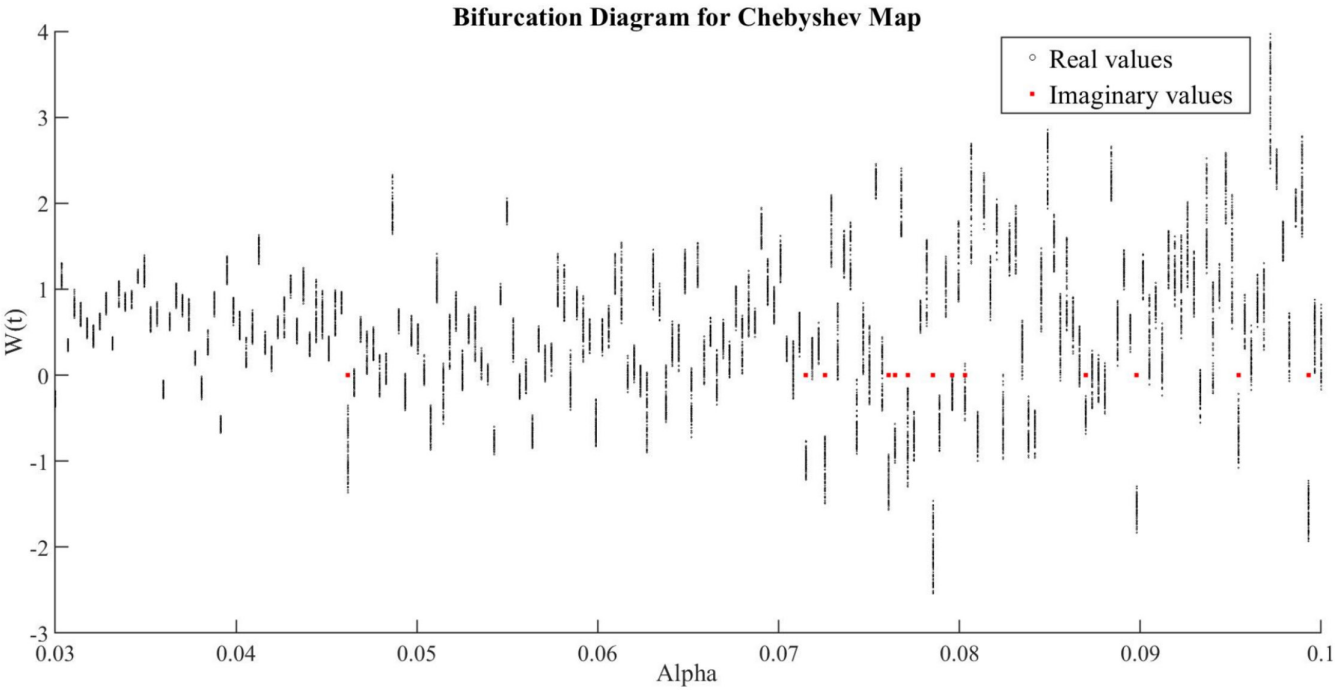



**FIGURE 9** | Bifurcation diagram of the chaotic hybrid system. Minor variations in parameters $\alpha$ and $\beta$ lead to significant changes in output, demonstrating exponential key sensitivity.

**TABLE 3** | Comparison of AES, standard chaotic methods, and the proposed hybrid model.

| Feature | AES | Standard chaos | Hybrid chaos (proposed) |
|---|---|---|---|
| Differential uniformity | 4.0 | 3–4 | **2–3** |
| Nonlinearity score | **12** | **12** | **12** |
| Avalanche effect | **0.50** | 0.52 | 0.53 |
| Key sensitivity | Medium | High | **Very high** |
| Computational complexity | Low | Medium | **Low** |

*Note:* Bold entries in this table highlight the most relevant feature(s) for each scheme (AES, standard chaos, hybrid chaos). Different features are emphasized depending on their importance in the comparison. This shows the strength of each scheme.

through chaotic injection and clips weight to prevent excessive growth. The final weight matrix using 100 neurons is shown in Figure 11 (a) and the entropy evaluation is shown in Figure 11b. Key space and run-time analyses are essential to evaluate the cryptographic strength and practical feasibility of the proposed model. Comparative analysis of entropy and key space expansion using Equations (13) and (14) indicates a 1.08% increase in bit entropy $H_{Bit}$ and a 6.6 times expansion in effective key space $E_K$ over standard $n$ byte AES.

$$\Delta H_{Bit}(\%) = \frac{H_{Bit}(Proposed) - H_{Bit}(AES)}{H_{Bit}(AES)} \times 100\% \tag{13}$$

$$E_K = 2^{n \cdot \Delta H_{\text{Bit}}} \tag{14}$$

The computational overhead $O(\%)$ introduced by the proposed modifications is minimal, measured at 3.2%, resulting in an encryption throughput $\mathscr{R}_{proposed}$ of 193.8 Mbps. The computational overhead and throughput are evaluated using Equations (15) and (16). Together, these results indicate that the proposed scheme achieves a substantial increase in key-space strength while maintaining efficient runtime performance suitable for real-world satellite communication scenarios.

$$O(\%) = \frac{T_{proposed} - T_{AES}}{T_{AES}} \times 100\% \tag{15}$$

$$\mathscr{R}_{proposed} = \frac{\mathscr{R}_{AES}}{1 + \frac{O}{100}} \tag{16}$$

The quantum-inspired encryption circuit demonstrates the role of chaotic perturbations in the transformation of plaintext states. The QFT block contributes to increased diffusion, making decryption infeasible without precise knowledge of the chaotic parameters. The circuit is defined as follows:

$$|q_0\rangle \rightarrow H \rightarrow X \rightarrow X^{\frac{1}{2}} \rightarrow QFT \rightarrow \text{Measure} \tag{17}$$

$$|q_1\rangle \rightarrow H \rightarrow Z \rightarrow Z^{\frac{1}{2}} \rightarrow QFT \rightarrow \text{Measure} \tag{18}$$

The robustness of the proposed chaotic injection with quantum wave collapse scheme against empirical ML attacks demonstrates a 36.8% absolute reduction in classifier success rate. The baseline AES exhibits an average classifier accuracy of 93.3%, whereas the proposed hybrid chaotic method reduces the attack accuracy to 55.5%. The average classifier accuracy, $\mathcal{A}$, is calculated using the number of ciphertext samples $N$, the ML



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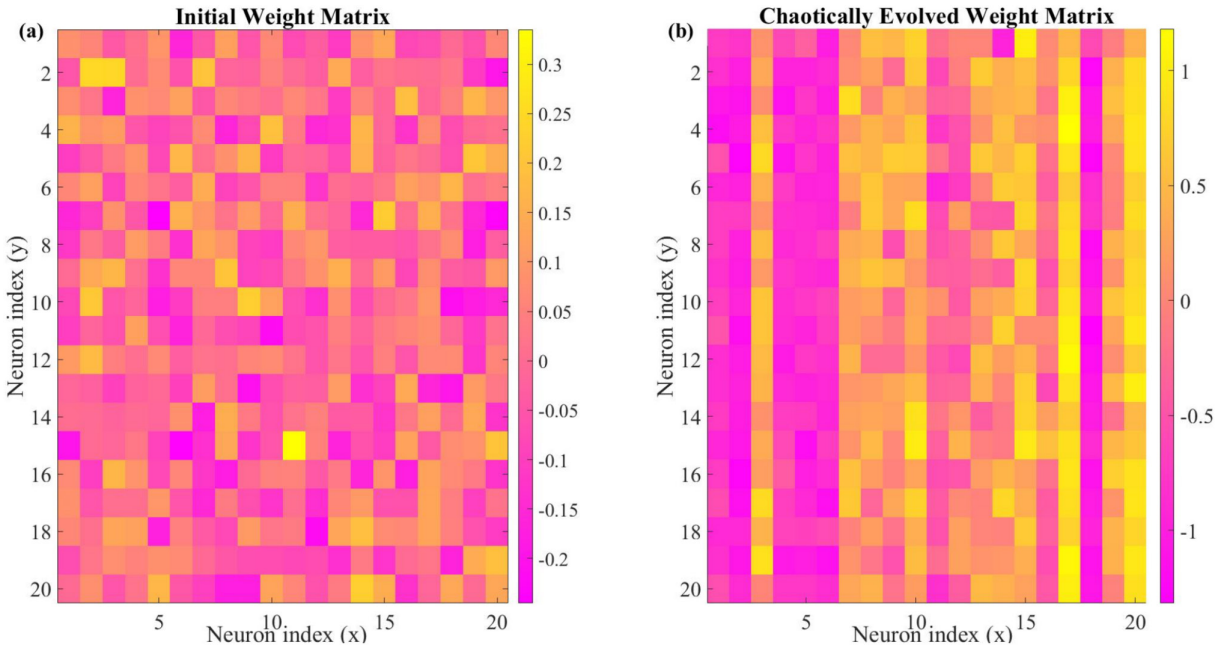



**FIGURE 10** | (a) Initial and (b) chaotically evolved weight matrices for neurons at different indices.

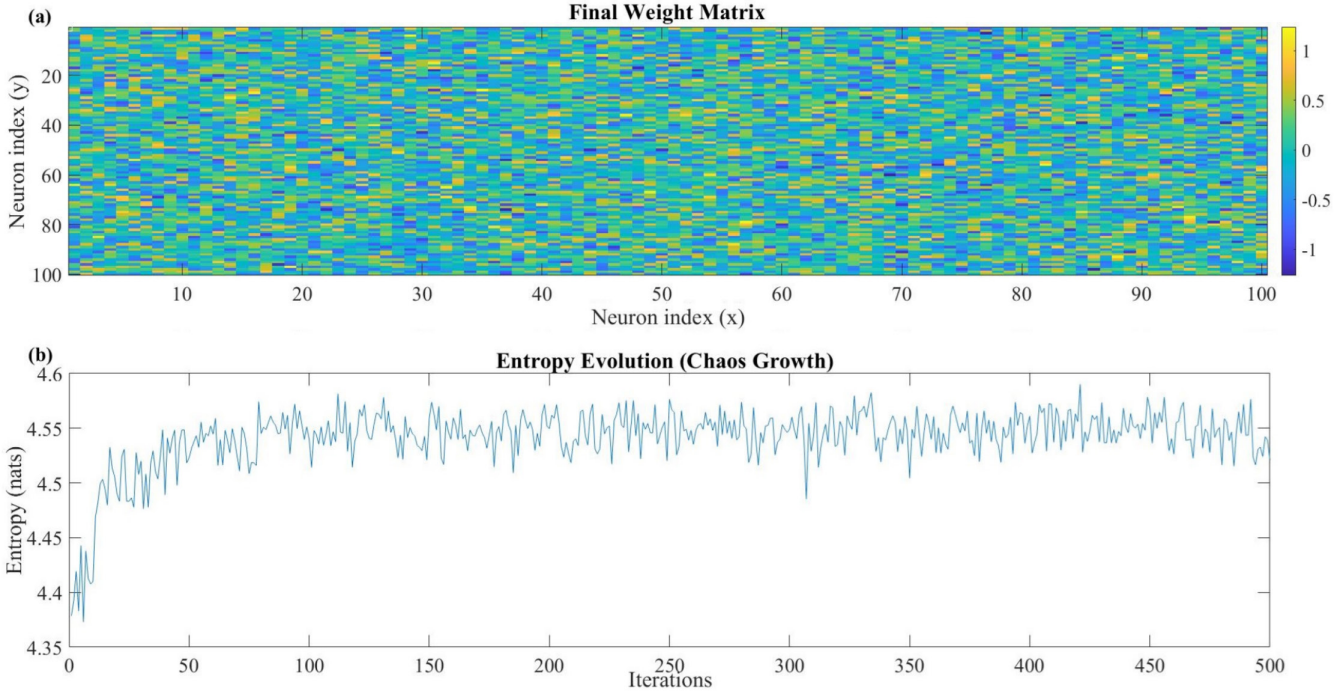



**FIGURE 11** | (a) Final weight matrix (100 neurons) and (b) entropy evaluation of chaotic growth shows steady-state entropy with chaotic fluctuations.

classifier $f_\theta(\cdot)$ with parameters $\theta$, the $i$th ciphertext $C_i$, and the ground-truth class label $k_i$ as shown in Equation (19).

$$\mathcal{A} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[f_\theta(C_i) = k_i\right] \tag{19}$$

The indicator function $\mathbf{1}[\cdot]$ equals 1 if the condition is true and 0 otherwise. A two-proportion $z$ test confirms that the reduction in classifier accuracy is statistically significant. The observed results are consistent with the information-theoretic limits, as the scheme effectively reduces the total-variation distance between class conditionals, which in turn tightens Fano's inequality bound on the minimum achievable classification error. The encryption shows a high sensitivity to initial conditions, making brute-force attacks impractical. The proposed method maintains a computational efficiency close to AES while improving cryptographic properties. The chaotic-quantum hybrid approach demonstrates improved resilience against statistical and differential cryptanalysis. The hybrid method achieves a balance



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between complexity and security, giving superior protection without excessive computations. The proposed model is efficient compared to AES. The results confirm that the proposed hybrid chaotic method strengthens encryption security through lower differential uniformity and improved randomness. Traditional AES S-Boxes follow deterministic substitution rules, making them vulnerable to algebraic attacks. In contrast, the chaotic S-Box introduces pseudo-random variations, making cryptanalysis significantly harder. Additionally, the quantum-inspired approach ensures that encrypted states remain probabilistically distributed, resembling quantum superposition. This property strengthens the model against adaptive chosen-ciphertext attacks (CCA). The integration of QFT further increases diffusion, preventing classical decryption attempts. To validate the claimed improvements, the proposed method is compared with recent state-of-the-art AES enhancement approaches. The triple-layer RGB image encryption algorithm [40] achieves a bit entropy of 7.999 and demonstrates improved key space expansion but suffers from lower resistance to differential attacks, reduced robustness against ML attacks, and higher computational overhead. In contrast, the proposed scheme achieves a larger effective key space and slightly higher bit entropy compared to the multi-chaotic key expansion method [41]. Recent hyper-chaotic schemes provide further enhancements. A memristive approach [31] achieves high security with a peak signal-to-noise ratio of 8.21 dB through unpredictability but incurs high computational cost, limiting practicality. Similarly, the multiple image encryption method [42] demonstrates strong metrics but remains constrained by computational costs. Specifically, the proposed approach improves resistance to differential attacks by approximately 25% while reducing the success rate of empirical machine learning attacks from 93.3% for baseline AES to 55.5%. In addition, computational overhead is kept low (3.2%), resulting in an encryption throughput of 193.8 Mbps, which is competitive with or superior to other AES-based hybrid schemes. Overall, this comparative evaluation confirms that the proposed method achieves improved cryptographic strength, maintains practical efficiency, and offers resilience against quantum attacks relative to recent works.

## 5 | Conclusions

The hybrid chaotic encryption method demonstrates strong security features while maintaining computational efficiency. By combining chaotic transformations with quantum-inspired processing, the model enhances resistance to differential and linear cryptanalysis. The integration with the AES establishes a robust framework for improving the security and reliability of satellite communication data transmission. Simulations validated its resilience, and the hybrid chaotic S-Box achieved improved differential uniformity while maintaining nonlinearity, strengthening AES as a reliable standard for sensitive information. However, the proposed model has some limitations. The hybrid chaotic S-Box, while improving cryptographic strength, shows a slightly higher avalanche effect, which may complicate performance tuning in practical systems. Additionally, the framework has only been validated in simulation. Real-world satellite environments with hardware constraints, latency issues, and channel noise can reveal new vulnerabilities or computational bottlenecks. The reliance on quantum-inspired mechanisms, though conceptually powerful, currently lacks hardware validation and may incur additional implementation costs. Finally, the scalability of the proposed method to large-scale data streams has not yet been tested, which may limit immediate adoption. Future extensions of this work will focus on addressing these limitations by exploring quantum error correction principles to enhance stability, optimizing avalanche properties, and conducting hardware-based validation. Extending this method to audio and video encryption will also provide valuable insights into performance and security constraints. Overall, this research lays the foundation for more secure and adaptive encryption techniques that merge quantum-inspired principles with classical cryptographic architectures while recognizing areas that require refinement before widespread deployment.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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